

JESSICA FINOCCHIARO

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APPOINTMENTS

Boston College

Sep 2024–Present

Tenure-Track Assistant Professor of Computer Science

Harvard University

Sep 2022–Aug 2024

National Science Foundation Mathematical Sciences Postdoctoral Research Fellow

Center for Research on Computation and Society Research Fellow

Host: Dr. Yiling Chen

EDUCATION

University of Colorado Boulder

Aug 2017 - Aug 2022

Ph.D., Department of Computer Science

M.S. Computer Science, May 2020

Advised by Dr. Rafael Frongillo

Florida Southern College (Honors College)

Aug 2013 - May 2017

Bachelor of Science; Computer Science, Mathematics.

Overall GPA: 4.0/4.0

Minor in Religion

KEYWORDS

Theoretical Machine Learning, Nonparametric Statistics, Surrogate Loss Design, Property Elicitation, Mechanism Design, Algorithmic Fairness

RESEARCH PUBLICATIONS

*($\alpha - \beta$) denotes alphabetical author ordering; * denotes equal contribution*

($x\%$) denotes the acceptance rate.

Conference proceedings listed above are considered full archival publications in Computer Science.

Analyzing surrogate loss design via $\mathcal{H}$ -calibration

JF, Sanket Shah, Milind Tambe

In submission to Conference on Learning Theory

Structured Prediction with Abstention via the Lovász Hinge

JF, Rafael Frongillo, Enrique Nueve IV ($\alpha - \beta$)

In submission to Journal of Machine Learning Research

Three Types of Calibration with Properties and their Semantic and Formal Relationships

Rabanus Derr, **JF**, Robert C. Williamson ($\alpha - \beta$)

Journal of Machine Learning Research

Robustness of voting mechanisms to external information

Yiling Chen, **JF** ($\alpha - \beta$)

Symposium on Algorithmic Game Theory 2025

Trading off consistency and dimensionality of convex surrogates for the mode

Enrique Nueve IV, Dhamma Kimpara, Bo Waggoner, **JF**

Neural Information Processing Systems (NeurIPS) 2024

Bridging Research and Practice Through Conversations with Practitioners

Mayra Russo, Mackenzie Jorgensen, Kristen M. Scott, Wendy Xu, Di H. Ngyuen, **JF**, Matthew Olckers

Equity and Access to Algorithms, Mechanisms, and Optimization (EAAMO) 2024

Using Property Elicitation to Understand Impacts of Fairness Constraints

JF

Fairness, Accountability, and Transparency (FAccT) 2024

Participatory Objective Design via Preference Elicitation

Ali Shirali, **JF**, Rediet Abebe

Fairness, Accountability, and Transparency (FAccT) 2024

Improving the Robustness of Predict-then-Optimize algorithms

Sonja Johnson-Yu, **JF**, Kai Wang, Arunesh Sinha, Yevgeniy Vorobeychik, Aparna Taneja, Milind Tambe

GameSec Conference on Decision and Game Theory 2023

Online Platforms and the Fair Exposure Problem Under Homophily

Jakob Schoeffler*, Alexander Ritchie*, Keziah Naggita*, Faidra Monachou*, **JF***, Marc Juarez

AAAI Conference on Artificial Intelligence 2023

An Embedding Framework for the Design and Analysis of Consistent Polyhedral Surrogates

JF, Rafael Frongillo, Bo Waggoner

Journal of Machine Learning Research

Supercedes NeurIPS 2019 paper

Lovász Hinge and The Structured Abstain Problem

JF, Rafael Frongillo, Enrique Nueve ($\alpha - \beta$)

Conference on Learning Theory (COLT) 2022

Consistent Polyhedral Surrogates for Top- k Classification and its Variants

JF, Rafael Frongillo, Emma Goodwill, Anish Thilagar ($\alpha - \beta$)

International Conference on Machine Learning (ICML) 2022

Unifying Lower Bounds on Prediction Dimension of Consistent Convex Surrogates

JF, Rafael Frongillo, Bo Waggoner

Neural Information Processing Systems (NeurIPS) 2021

A Decomposition of a Complete Graph with a Hole

Roxanne Back, Alejandra Brewer Castano, Rachel Galindo, **JF**

Open Journal of Discrete Mathematics Vol. 11 No. 1. (2021)

Bridging Machine Learning and Mechanism Design towards Algorithmic Fairness

JF, Roland Maio, Faidra Monachou, Gourab Patro, Manish Raghavan, Ana-Andreea Stoica, Stratis Tsirtis ($\alpha - \beta$)

Fairness, Accountability, and Transparency (FAccT) 2021

Evolutionary Optimization of Cooperative Strategies for the Iterated Prisoner's Dilemma

JF, H. David Mathias

IEEE Transactions on Games 2020. Supersedes GECCO 2017 paper.

Embedding Dimension of Polyhedral Losses

JF, Rafael Frongillo, Bo Waggoner

Conference on Learning Theory (COLT) 2020

An Embedding Framework for Consistent Polyhedral Surrogates

JF, Rafael Frongillo, Bo Waggoner

Neural Information Processing Systems (NeurIPS) 2019

Convex Elicitation of Continuous Real Properties

JF, Rafael Frongillo

Neural Information Processing Systems (NeurIPS) 2018

Spotlight acceptance; about 4% of submitted papers

Social trends in the Iterated Prisoner's Dilemma

JF, H. David Mathias

Genetic and Evolutionary Computation Conference (GECCO) 2017

Egocentric Height Estimation

JF, Aisha Urooj Khan, Ali Borji

Winter Conference on Applications in Computer Vision (WACV) 2017 (44%)

TEACHING EXPERIENCE

Data Science Principles (Boston College CSCI 1090)

Aug–Dec 2025

Professor

Randomness and Computation (Boston College CSCI 2244)

Jan–May 2025

Professor

Foundations of Algorithmic (Un)Fairness (Boston College CSCI 3358)

Aug–Dec 2024

Professor

AWARDS AND FUNDING

2025 Schiller Institute SI-RITEA Type A Grant co-PI: Francisco Marmolejo-Cossío (\$15,600)

2024 Schiller Institute Research Funding (\$2,500)

2022 National Science Foundation Mathematical Sciences Postdoctoral Research Fellowship (\$150,000)

2021-22 CU Boulder Outstanding Research Award (\$600)

2021-22 CU Boulder Research Expo Poster Session Winner (\$500)

2021 CU Boulder Women who makes a difference

2021 CU Boulder Endowed Founder's Fund Recipient (\$500)

2021, 2020 CU Boulder Publication Award (\$200)

2020 Rising Stars in EECS at University of California Berkeley

2019-2021 National Science Foundation Graduate Research Fellowship (16% grant rate; \$102,000)

2018, 2019 Travel Grant for the ACM Economics and Computation Conference's Algorithmic Game Theory Mentoring Workshop (\$500)

2018-19, 2019-20, 2021-22 University of Colorado Boulder Outstanding Student Service Award

2017 Student travel grant to attend the Joint Meetings in Mathematics from the Mathematical Association of America

2017 Florida Southern College Honor Walk Graduate (Most Outstanding Senior)

2017 Florida Southern College Senior Math Student of the Year

2016, 2017 Florida Southern College Female Resident Advisor of the Year

INVITED TALKS

“Urban-Rural Disparities in FEMA’s National Risk Index”

- University of Colorado Meet-a-Data-Scientist Workshop, January 20, 2026

“Hierarchies of Calibration Definitions for Multiclass Settings”

- Boston Economics and Computation Hub Workshop, November 18, 2025

“Property elicitation and Loss function choice”

- Boston College Experimental Math and Machine Learning seminar, April 18, 2025

“Introduction to Computational Social Choice”

- Public Interest Technology New England Summer Impact Fellowship Program, June 18, 2024

“Introduction to EAAMO methods”

- Public Interest Technology New England Summer Impact Fellowship Program, June 4, 2024

“Visualizing our values: using property elicitation to understand the impacts of fairness constraints”

- Human Machine Decision-Making reading group, MIT, September 26, 2023
- EC Gender Inclusion Workshop, July 11, 2023

“Designing convex surrogates for discrete prediction tasks via embeddings”

- Microsoft Research Seminar September 10, 2023
- Peking University April 12, 2023
- Boston University CS Theory Seminar, April 10, 2023
- University of Massachusetts CS Theory Seminar March 28, 2023
- Colorado School of Mines, November 28, 2022
- Brown University, November 16, 2021
- CU Boulder Statistics, Optimization, and Machine Learning Seminar (StatOptML), Nov 2, 2021
- CU Boulder Algorithmic Economics Reading Group, May 10, 2021 [Recording](#)

“Introduction to Algorithms and Machine Learning” Morehouse College Workshop on Algorithms and Society, May 2022

“Fairness and Discrimination Through the Dual Lens of Mechanism Design and Machine Learning”

- Brown University Data Science Institute Fairness Fest, Feb 24, 2022
- Tutorial at ACM Conference on Economics and Computation (EC) 2021
- Princeton Technology for a Just Society (JuST) Speaker Series, October 14, 2020

“An Introduction to Property Elicitation and Connections to Machine Learning”

- Brown University, Nov 9, 2021
- Boston College Experimental Math and Machine Learning Seminar April 26, 2025

“Indirect Elicitation as a Necessary Condition for Consistency” RIKEN AIP Learning Theory Seminar, December 14, 2020

WORK EXPERIENCE

People, Technology & Processes, LLC

Aug 2015–May 2016

Software Engineering Intern

- Developed a backend in C# and R for visually analyzing ticket sales in Tableau for various NFL, NBA, NHL, and MLB teams

LEADERSHIP

Equity and Access to Algorithms, Mechanisms, and Optimization

March 2018-July 2023

Community Engagement Co-lead Aug 2021–July 2023

Co-Organizer for the working group on Discrimination and Equality in Algorithmic Decision-making Jan–June 2021

Co-Organizer for the working group on Bias and Fairness Feb–Dec 2020

Representative at Public Interest Technology New England Regional Meetings

CU Boulder Computer Science Graduate Student Association

January 2018–December 2020

Vice-Chair 2018-19, Chair 2020

Co-founded the organization in January 2018 to advocate for student well-being in the department and facilitate communication between students and faculty

Worked with department to advocate for student needs and well-being on issues regarding course requirements, advising, and social events

Helped organize the PhD Open House recruitment event for ~70 prospective PhD students in February 2019 and 2020

Worked with department to establish opportunities for students to share about their culture through both recurring and annual events

Student member of the 2022 faculty search committee

ACADEMIC SERVICE

Postdoctoral Supervision

- **Maneesha Papireddygar**, co-advised with Francisco Marmolejo-Cossío

Undergraduate Research Advising

- **Tara Balan** *Evaluating the efficacy of Olympic team selection mechanisms*, Aug 2024 – May 2025.
- **Patrick Lanza** *Testing and measuring multiclass Γ -calibration*, June 2025 – August 2025.
- **Marcus Rothera** *Tradeoffs in ML model size and loss function choice*, June 2025 – present.
- **Andrew Ren** *Double descent with general loss functions*, June 2025 – present.
- **Sarah Fleming** *Designing smooth loss functions for discrete decision tasks*, June 2025 – present.

- **Kit Levy** *Modeling the effect of voter turnout on social welfare, June 2025 – present.*
- **Andy Bello** *Attribute-based elicitation for matching, June 2025 – present.*

Boston College Internal Service

- Women in Computer Science (WiCS) faculty advisor (*Fall 2024 – present*)
- McNair Exploratory Program Mentor (*Fall 2024 – present*)

Mentorship programs

- Public Interest Technology New England Rising Stars Program Mentor 2024
- Economics and Computation Mentoring Workshop Faculty Market Panelist 2024
- MD4SG Mentorship Program 2022
- LatinX in AI 2021
- Black in AI 2019-21
- CU Boulder United Government of Graduate Students Peer Mentoring 2019-20
- Organized and matched students for [CU Boulder Computer Science Peer Mentoring 2020](#)

Reviewing and Metareviewing

- Metareviewing
 - Equity and Access to Algorithms, Mechanisms, and Optimization (EAAMO) 2025
 - Fairness, Accountability and Transparency (FAccT) 2025, 2026
- Reviewing
 - Journal of AI Research 2025
 - Association for the Advancement of Artificial Intelligence (AAAI) 2025
 - Conference on Learning Theory (COLT) 2025
 - Web and Internet Economics (WINE) 2023, 2024, 2025
 - Economics and Computation (EC) 2023, 2026
 - IJCAI Workshop on Computational Fair Division 2023, 2024
 - Autonomous Agents for Social Good workshop 2024, 2025
 - Symposium on Discrete Algorithms (SODA) 2023
 - Uncertainty in Artificial Intelligence (UAI) 2023
 - Electronic Journal of Statistics 2022
 - Fairness, Accountability, and Transparency (FAccT) 2022, 2023
 - Equity and Access in Algorithms, Mechanisms, and Optimization (EAAMO) 2021, 2022, 2023, 2024
 - International Conference on Machine Learning (ICML) 2021, 2022
 - Harvard Center for Research on Computation and Society Workshop on AI for Social Good 2020
 - Mechanism Design for Social Good (MD4SG) Workshop 2020
 - Neural Information Processing Systems (NeurIPS) 2020, 2021
 - Women in Machine Learning (WiML) 2019

Workshop and Conference Organization

- Equity and Access in Algorithms, Mechanisms, and Optimization (EAAMO) 2023 Social co-chair
- Equity and Access in Algorithms, Mechanisms, and Optimization (EAAMO) 2022 Proceedings co-chair
- Equity and Access in Algorithms, Mechanisms, and Optimization (EAAMO) 2021 Posters co-chair
- Mechanism Design for Social Good (MD4SG) Workshop 2020 Poster Session co-chair

Conference Volunteering

- NeurIPS 2019, 2020
- Economics and Computation (EC) 2019, 2020
- WiML 2019

Neural Network

Conducted and published [interviews](#) of CU Boulder Computer Science students, faculty, and staff